Unit 1 – Using Objects and Methods
Checkpoint Practice 3

Name: **KEY**

1) Write what the following code segments will evaluate to.

```
String str = "Let's practice some";
```

Code	Evaluates To
`str.substring(6, 14)`	`practice`
`str.substring(0,2) + str.substring(15)`	`Lesome`
`str.indexOf(" ")`	`5`
`str.substring(16).length()`	`3`
`str.substring(str.length() - 7)`	`ce some`
`str.substring(str.substring(2, 11).length())`	`ctice some`

2) Write what the following code segments will evaluate to.

```
String phrase = "hello world!";
```

Code	Evaluates To
`phrase.substring(2)`	`llo world`
`phrase.substring(phrase.length()-2)`	`d!`
`phrase.substring(0, 5) + phrase.substring(6)`	`helloworld!`
`phrase.substring(phrase.length()-4, phrase.length()-3)`	`r`

3) Write what the following code segments will evaluate to.

```
int x = 5;
double y = 2.5;
```

Code	Evaluates To
`x + Math.pow(x, 2) % 3 + (int) y`	`8.0`
`(int)(Math.pow(y, 2) + 0.5) % 4`	`2`
`x / y + Math.abs(y - x) + (int) Math.sqrt(x * 4)`	`8.5`
`Math.abs(y - Math.pow(Math.sqrt(16), 2))`	`13.5`

4) Write a Java expression to find $\left(\sqrt{64}+3\right)^2$

```
Math.pow((Math.sqrt(64) + 3), 2)
```

5) Write an expression that calculates the absolute value of −15 then adds it to the square root of 121.

```
Math.abs(-15) + Math.sqrt(121)
```

6) Write an expression that generates a random integer from 10 to 100.

```
(int)(Math.random() * 91) + 10
```

7) Write an expression that finds the absolute difference between 12 and 25 raised to the power of 2.

```
Math.pow(Math.abs(12 - 15), 2)
```

8) Write an expression that takes the square root of 36 and multiplies it by a random integer between 1 and 5.

```
Math.sqrt(36) * ((int)(Math.random() *  5) + 1)
```

9) For the following code segments, determine if a positive number, negative number, or 0 is printed and explain why.

Code Segment	Number Printed and Why

`String word1 = "Apple";`
`String word2 = "Banana";`
`System.out.println(word1.compareTo(word2));`
	negative number since Apple comes before Banana alphabetically

`String word1 = "Zebra";`
`String word2 = "Monkey";`
`System.out.println(word1.compareTo(word2));`
	positive number since Zebra comes after Monkey

`String word1 = "Hello";`
`String word2 = "Hello";`
`System.out.println(word1.compareTo(word2));`
	0 since the words are identical

`String word1 = "hello";`
`String word2 = "Hello";`
`System.out.println(word1.compareTo(word2));`
	positive number since uppercase letters have lower ASCII values than lowercase letters, so hello is “greater” than Hello (note: not tested on AP exam)

`String word1 = "Program";`
`String word2 = "Programming";`
`System.out.println(word1.compareTo(word2));`
	negative number since Programming is longer but starts the same as Program, it is considered “greater”. (note: not tested on AP exam)

C 10) Consider the following method, which is intended to return true if at least one of the three strings `s1`, `s2`, or `s3`, contains the substring "no", otherwise the method should return false.

```
public boolean containsNo(String s1, String s2, String s3)
{
     String together = s1 + s2 + s3;
     return (together.indexOf("no") != -1);
}
```

Which of the following method calls (located in the same class) demonstrates that the method does **not** work as intended?

(A) `containsNo("note", "ignore", "nomad");`
(B) `containsNo("laughing", "happy", "noteworthy");`
(C) `containsNo("solution", "over", "loving");`
(D) `containsNo("enough", "button", "octopus");`

(C) should return false, because there are no occurrences of "no" in any of the words, but it will return true because when solution and over get put together, you will get a "no"

C 11) Which of the following correctly gives random integers between –7 and 2 inclusive?

(A) `int n = (int)(Math.random() * 8) - 7;`
(B) `int n = (int)(Math.random() * 9) - 7;`
(C) `int n = (int)(Math.random() * 10) - 7;`
(D) `int n = (int)(Math.random() * 11) - 7;`

2 – (-7) = 9 + 1 = 10 possible values for the range

-7 as the starting value

D 12) Consider the following code:

```
String word1 = "butter";
String word2 = "letter";
System.out.println(word1.indexOf(word2.substring(3)));
```

What is the output?

(A) –1
(B) 1
(C) 2
(D) 3

word2.substring(3) → ter

word1.indexOf("ter") = 3

D 13) If the following returns a positive number, what must be true about `str1` and `str2`?

```
str1.compareTo(str2)
```

(A) `str1` comes before `str2` alphabetically
(B) `str1` is longer than `str2`
(C) `str1` equals `str2`
(D) `str1` comes after `str2` alphabetically